

→ sets that are number systems:-

$$\Rightarrow \mathbb{N}, \mathbb{Z}, \mathbb{R}$$

$$A = \{a, b, c\}$$

→ addition or multiplication gives a new element.

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$$10 + 11 = 21 \\ 21 \in \mathbb{N}$$

→ $\mathbb{N}$ is closed under addition and multiplication operations.

$$4 \cdot 3 = 12 \\ 12 \in \mathbb{N}$$

→ commutative

$$2 + 4 = 4 + 2$$

→ associative

$$2 + 3 + 4 = (2 + 3) + 4 \\ 2 + (3 + 4)$$

→ distributive

$$2 \cdot (3 + 4) = 2 \cdot 3 + 2 \cdot 4$$

✓
→ natural ordering

$$6 < 7$$

$\mathbb{N}$
 $\mathbb{Z}$
 $\mathbb{R}$

$$\frac{1}{10^{1000+2}}$$

→ well-ordering principle

$$A = \{5, 6, 7, 8, 9, 10, \dots, 100\}$$

$$\frac{1}{n} > \frac{1}{n+1}$$

$$\rightarrow A = \left\{ \frac{1}{n} ; n \in \mathbb{N} \right\}$$

A well-ordering...

$$A = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{1001}, \frac{1}{1002}, \dots \right\}$$

does not apply!